



Intelligent optimal design of steel-concrete hybrid wind turbine tower based on evolutionary algorithm

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ARTICLE INFO

Keywords:

Wind turbine
Steel-concrete hybrid tower
Evolutionary algorithm
Structural optimization
Intelligent optimal design
FE analysis

ABSTRACT

The steel-concrete hybrid wind turbine tower possesses the advantages of high stiffness and low comprehensive cost, showing promising prospects in applying tall wind turbine towers. However, the present design of hybrid towers faces challenges such as low work efficiency and a high degree of repetitive labor. In this study, an intelligent optimization framework for hybrid tower structures has been developed based on parametric finite element (FE) analysis and evolutionary algorithms. The tower radii, thicknesses, height of concrete segment, and prestressed steel strands area are selected as optimization variables. Firstly, an optimization mathematical model for the hybrid tower was proposed considering a series of design constraints and construction factors. A refined hybrid tower FE model using the OpenSees program was established, and a parametric modeling process was implemented to evaluate the constraints during the optimization process. Then, an efficient optimization algorithm applicable to the hybrid tower was developed based on modified evolutionary algorithms: genetic algorithm and differential evolution algorithm. Furthermore, a graphical user interface (GUI) connecting different functional modules of the optimization mathematical model, FE analysis model, and intelligent optimization algorithm was presented. Finally, taking a 5 MW steel-concrete hybrid tower with a hub height of 160 m as a case study, the total construction cost of intelligent optimization design is reduced by about 11% compared with the conventional manual optimization design while satisfying all design constraints. The results indicate that the proposed intelligent optimization design framework can effectively and accurately provide the excellent design of the hybrid tower structure.

1. Introduction

In the face of common challenges such as energy shortages, environmental pollution, and climate change, a global energy revolution has emerged focusing on the vigorous development and utilization of renewable energy. Wind energy, characterized by abundant resources, widespread distribution, and cleanliness, plays a crucial role in effectively alleviating current energy shortages and environmental pollution caused by the irrational use of traditional fuels [1–3]. In the long term, wind energy resources have significant potential for large-scale development and commercialization. In recent decades, wind turbines have experienced rapid growth and application.

According to the Global Wind Energy Council (GWEC) report for 2023, the global newly installed wind power capacity reached 77.6 GW in 2022, with onshore wind power installations accounting for 69 GW

and offshore wind power installations accounting for 9 GW [4]. Although offshore wind turbines yield greater electricity output and demonstrate enhanced operational stability, the associated costs are notably higher due to challenging environmental conditions and intricate load factors. Consequently, onshore wind turbines currently dominate the primary market and exhibit substantial development potential. To achieve more significant economic benefits, the trend in the wind turbine industry is towards using high-capacity wind turbines and ultra-high hub heights, which also places higher demands on the structural performance of wind turbine towers. The most widely used single-tube steel tower is designed to meet bearing capacity and stiffness by increasing the diameter and thickness (Fig. 1(a)), which increases the difficulty of transportation and processing and raises the cost of tower construction [5]. Steel-concrete hybrid tower is a novel tower structure with a lower section comprising precast concrete tower segments and an

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<https://doi.org/10.1016/j.jcsr.2024.108729>

Received 17 January 2024; Received in revised form 30 March 2024; Accepted 27 April 2024

Available online 7 May 2024

0143-974X/© 2024 Published by Elsevier Ltd.

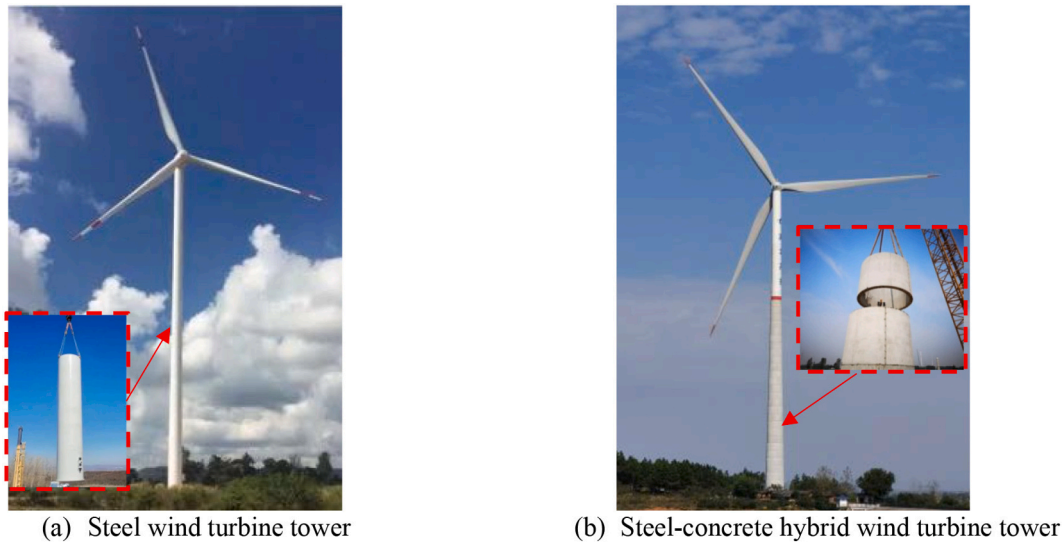


Fig. 1. Main types of wind turbine towers.

upper section comprising steel tower segments (Fig. 1(b)). The advantages of precast concrete tower segments include high production efficiency, convenient transportation in sections, and rapid assembly and hoisting. Externally prestressed steel strands are also added to the concrete tower segments to avoid tensile stress along the height of the concrete tower segment, ensuring that the entire cross-section of the concrete is under compression and preventing concrete from cracking under tension, thereby meeting the safety and applicability of the entire structural system. The steel-concrete hybrid tower exhibits notable advantages, including high overall stiffness and low total cost, thereby presenting promising application prospects for wind turbines with hub heights exceeding 120 m [6].

The wind turbine tower cost typically accounts for 20%–30% of the total wind turbine system cost, and often, the same tower design is employed for identical turbine models within the same wind farm [7,8]. Therefore, optimizing the tower structure design can improve the economic benefits and shorten the investment payback period of wind turbine systems. However, the current design of wind turbine tower structures faces challenges such as low work efficiency, repetitive labor, and dependence on individual experience of engineers, making it challenging to obtain the optimal structural solution. Conventional design method is highly influenced by subjective factors and is time-consuming, making it unable to meet the demands of the rapidly evolving wind turbine industry. Implementing intelligent methods for automated modeling, analysis, and optimization design of structures can effectively address the inefficiencies associated with conventional wind turbine tower structural design, ensuring the economic viability of the optimized structure.

Recently, many scholars have explored and studied intelligent optimization designs for wind turbine towers. Uys et al. [9] used the Rosenbrock direct search method to optimize the shell thickness, stiffeners number, and stiffeners dimensions for steel towers, with the total cost of materials and manufacturing as the objective function. Karpat [10] developed software for cost optimization of steel towers in stiffened wind turbines based on particle swarm optimization algorithm and MATLAB software, taking into account factors such as materials, forming, welding, coating, and cutting in the cost. Yoshida [11] used genetic algorithms to optimize the design of traditional steel tower structures by constructing penalty functions to set constraint conditions. However, specific design details for wind turbine towers were not introduced in references [9–11]. Al-Sanad [12,13] proposed a structural optimization framework for onshore steel towers based on the FE software ANSYS and genetic algorithms. Ma [14] and Lana et al. [15] used computer-aided

engineering (CAE) tools and genetic algorithms to optimize circular and octagonal pre-stressed concrete towers with a height of 100 m, respectively. Sophisticated finite element software has been adopted for optimization in references [12–15], resulting in low computational efficiency.

Compared with traditional steel towers, the design of steel-concrete hybrid wind turbine towers involves more design variables and constraints, and the mechanical analysis of hybrid wind turbine towers is more complex. Conducting multi-variable optimization design for hybrid towers to reduce costs is of significant importance and engineering value for promoting the widespread application of hybrid towers. Chen et al. [16] optimized the steel-concrete hybrid tower based on a modified genetic algorithm, and the results showed that the optimized tower exhibited good tower stiffness and economic performance. However, fatigue and natural frequency constraints were not considered in the optimization process, which may lead to ineffective optimization in structures where fatigue and natural frequency play a crucial control role. Xu et al. [17] employed a modified particle swarm optimization algorithm to optimize the steel-concrete hybrid tower, and the results demonstrated a cost reduction of approximately 27% compared to traditional steel towers. However, they did not provide a method for calculating the frequency of the hybrid tower, and the concrete fatigue calculation method provided was not applicable to high-cycle fatigue structures such as wind turbines. Huang et al. [18] explored the correlation between the natural frequency and geometric shape of the steel-concrete hybrid tower through sensitivity analysis, and applied genetic algorithm to geometric optimization of the 160 m high hybrid tower, effectively reducing the total cost. However, the author did not consider the influence of the rotational inertia of the top wind turbines when deriving the frequency calculation formula, and the analysis of prestress as an external load does not conform to the actual loading conditions. Additionally, only simplified calculation methods are used for fatigue calculation. References [16–18] failed to effectively address the constraints of natural frequency and fatigue design in the optimization of hybrid towers. While some researchers have paid attention to the optimization design of hybrid towers, there are still some shortcomings in mechanical analysis, computational efficiency, and optimization methods. An FE analysis model was developed for hybrid tower based on the open-source FE program OpenSees in this study to address the mentioned issues: 1) Compared with solid FE models, the beam-column model was adopted to improve computational efficiency; 2) The rotational inertia of wind turbines and the interaction between prestressed steel strands and concrete tower are considered to improve

calculation accuracy; 3) The influence of geometric nonlinear effects is also considered in the model, as the wind turbine tower is a highly flexible structure; 4) OpenSees provides greater flexibility in software development permissions, facilitating integration with intelligent optimization algorithms.

This paper is organized as follows. In Section 2, a nonlinear multi-constraint optimization mathematical model for steel-concrete hybrid towers is proposed. In Section 3, a refined FE model for a hybrid tower is developed based on the open-source FE program OpenSees. Section 4 presents modified genetic algorithm and modified differential evolution algorithm suitable for optimizing hybrid towers and introduces the intelligent optimization design framework for hybrid towers. Besides, the practicability and efficiency of the intelligent optimization design framework proposed in this paper were verified using the actual engineering case of a 5 MW hybrid tower.

2. Mathematical model for hybrid tower optimization

The optimization problem of hybrid tower involves multiple variables and constraints, which can be generalized as the problem of finding the minimum value of a function subject to specified constraints. The mathematical model for this can be expressed as follows:

$$\begin{aligned} &\text{find : } x = \{x_1, x_2, \dots, x_n\} \\ &\text{minimize : } f(x) \\ &\text{subject to : } g_i(x) \leq 0 \quad (i = 1, 2, \dots, m) \\ &\quad x_{\min} \leq x \leq x_{\max} \end{aligned} \quad (1)$$

where x represents a vector of hybrid tower design variables; $f(x)$ denotes the objective function; and $g_i(x)$ represents the optimization constraints.

2.1. Design variables

This optimization problem for hybrid tower involves a total of 9 design variables, namely the bottom radius (r_{cb}) and thickness (t_{cb}) of the concrete tower segment, the top radius (r_{ct}) and thickness (t_{ct}) of the concrete tower segment, the bottom radius (r_{sb}) and thickness (t_{sb}) of the steel tower segment, the top radius (r_{st}) and thickness (t_{st}) of the steel tower segment, and the height (H_c) of the concrete tower segment.

2.2. Design constraints

The hybrid tower should meet the design requirements for natural frequency, ultimate limit state, serviceability limit state, and fatigue. Additionally, design variables must comply with constraints on geometric dimensions. The tower design constraints based on the above requirements can be expressed as follows, where g_1 - g_{11} represent performance constraints, and g_{12} - g_{16} represent geometric constraints.

2.2.1. Natural frequency constraint

To avoid resonance between the tower and the wind turbine, the first-order frequency of the tower should steer clear of the $\pm 10\%$ range of the wind turbine rotational frequency and the blade passing frequency [19]. Therefore, three control strategies can be employed: a soft design scheme with tower fundamental frequency below $0.9f_{1p}$; a stiff-soft design scheme with tower fundamental frequency between $1.1f_{1p}$ and $0.9f_{3p}$; and a stiff design scheme with tower fundamental frequency exceeding $1.1f_{3p}$. For steel-concrete hybrid towers, the stiff-soft design scheme tends to be more economical [6]. The constraint conditions for natural frequency are expressed as follows using the function $\text{sgn}(x)$:

$$g_1 = \text{sgn}(f_{1st} - 1.1f_{1p}) \cdot \text{sgn}(f_{1st} - 0.9f_{3p}) \leq 0 \quad (2)$$

where f_{1st} is the first-order frequency of the tower; f_{1p} and f_{3p} turbine rotational frequency and the blade passing frequency, respectively.

2.2.2. Ultimate limit state constraint

Under the ultimate limit state: (1) The maximum stress in the steel tower segment should not exceed the design strength of steel. (2) The maximum compressive stress and tensile stress in the concrete tower segment should not exceed the design compressive and tensile strength of concrete, respectively. (3) Local stability of the steel tower segment is evaluated by Eq. (6) according to Chinese standard GB 50135-2019 [20]. (4) Overall stability of the steel tower segment is evaluated by Eq. (7) according to Eurocode 3 BS EN 1993-1-6 [21].

$$g_2 = \sigma_{s,max}^u - [f] \leq 0 \quad (3)$$

$$g_3 = \sigma_{c,max}^u - [f_c] \leq 0 \quad (4)$$

$$g_4 = \sigma_{c,tmax}^u - [f_t] \leq 0 \quad (5)$$

$$g_5 = \sigma_{scr,max}^u - [\sigma_{cr}] \leq 0 \quad (6)$$

$$g_6 = \left(\frac{\sigma_{x,Ed}}{\sigma_{x,Rd}} \right)^{k_x} + \left(\frac{\tau_{x\theta,Ed}}{\tau_{x\theta,Rd}} \right)^{k_\tau} - 1 \leq 0 \quad (7)$$

where $\sigma_{s,max}$, $\sigma_{c,max}$, $\sigma_{c,tmax}$ and $\sigma_{scr,max}$ are maximum stress of steel, maximum compressive stress of concrete, maximum tensile stress of concrete, and maximum local stable critical stress of steel, respectively. $[f]$, $[f_c]$, $[f_t]$ and $[\sigma_{cr}]$ are the design values for steel strength, concrete compressive strength, concrete tensile strength, and steel local stable critical stress, respectively. $\sigma_{x,Ed}$ and $\tau_{x\theta,Ed}$ are the design values of the acting buckling-relevant meridional membrane stress and circumferential membrane and stress, respectively. $\sigma_{x,Rd}$ and $\tau_{x\theta,Rd}$ are the meridional design buckling stress and shear design buckling stress, respectively. k_x and k_τ are the buckling interaction parameters.

2.2.3. Serviceability limit state constraint

Under serviceability limit state: (1) The concrete tower segment is not allowed to experience tensile stress. (2) The maximum displacement at the top of the tower should not exceed $1/100$ of the total tower height to prevent excessive vibrations [20]. (3) The maximum rotation angle at the top of the tower should not exceed 5° to prevent the blades from contacting the tower due to tower bending [22].

$$g_7 = \sigma_{c,tmax}^s \leq 0 \quad (8)$$

$$g_8 = \Delta_{max} - H/100 \leq 0 \quad (9)$$

$$g_9 = \theta_{max} - 5 \leq 0 \quad (10)$$

where $\sigma_{c,tmax}^s$ is the maximum tensile stress in the concrete tower segment; H is the total tower height; Δ_{max} and θ_{max} are the maximum displacement and maximum rotation angle at the top of the tower, respectively.

2.2.4. Fatigue constraint

Due to the fluctuating wind loads, the tower structure experiences alternating stress states throughout its entire service life. The issue of fatigue damage in the tower structure cannot be ignored. By applying the Markov fatigue load matrix at the top of the tower, stress spectra at any point on the tower can be obtained, including stress amplitude levels $\Delta\sigma_1, \Delta\sigma_2, \dots, \Delta\sigma_i$, and the corresponding number of cycles $n_1, n_2, \dots, n_i$ at that point. The cumulative damage for steel and concrete is calculated based on the classical Palmgren-Miner linear damage accumulation theory.

Referring to Eurocode 3 BS EN 1993-1-9 [23] for fatigue strength analysis of steel, when the cumulative damage D_s for steel does not exceed 1, the fatigue strength is considered sufficient.

Table 1

Design variables for hybrid tower structure.

Design variable parameters	Symbols	Min value	Max value
Height of concrete tower segment (mm)	H_c	3100	151,900
Radius at concrete tower segment base (mm)	r_{cb}	2500	5000
Radius at concrete tower segment top (mm)	r_{ct}	1500	2000
Thickness at concrete tower segment base (mm)	t_{cb}	180	400
Thickness at concrete tower segment top (mm)	t_{ct}	180	400
Radius at steel tower segment base (mm)	r_{sb}	1500	2500
Radius at steel tower segment top (mm)	r_{st}	1500	2500
Thickness at steel tower segment base (mm)	t_{sb}	10	40
Thickness at steel tower segment top (mm)	t_{st}	10	40
Area of prestressed steel strands (mm ²)	A_s	3000	6000

$$g_{10} = D_s = \sum_{i=1}^j \frac{n_{si}}{N_{si}} - 1 \leq 0 \quad (11)$$

where n_{si} and N_{si} are the number of cycles and fatigue limit of the steel material corresponding to $\Delta\sigma_i$, respectively.

According to CEB-FIP guidelines [24] for fatigue strength analysis of concrete, the fatigue limit of concrete N is calculated by Eqs. (12–17).

$$S_{cd,min} = \gamma_{sd} \cdot \sigma_{c,min} / f_{cd,fat} \quad (12)$$

$$S_{cd,max} = \gamma_{sd} \cdot \sigma_{c,max} / f_{cd,fat} \quad (13)$$

$$\lg N_1 = \frac{8}{Y-1} \cdot (S_{cd,max} - 1) \quad (14)$$

$$\lg N_2 = 8 + \frac{8 \ln(10)}{Y-1} \cdot (Y - S_{cd,min}) \cdot \lg \left(\frac{S_{cd,max} - S_{cd,min}}{Y - S_{cd,min}} \right) \quad (15)$$

$$Y = \frac{0.45 + 1.8 S_{cd,min}}{1 + 1.8 S_{cd,min} - 0.31 S_{cd,min}^2} \quad (16)$$

$$N = \begin{cases} N_1 & \lg N_1 \leq 8 \\ N_2 & \lg N_1 > 8 \end{cases} \quad (17)$$

where $S_{cd,min}$ and $S_{cd,max}$ are the minimum and maximum compressive stress level, respectively. $\sigma_{c,min}$ and $\sigma_{c,max}$ are the minimum and maximum compressive stresses calculated for concrete, respectively. $f_{cd,fat}$ is the fatigue reference compressive strength. γ_{sd} is the partial factor. The fatigue strength is considered sufficient when the cumulative damage D_c for concrete does not exceed 1. Eq. (18) is adopted to check whether fatigue failure occurs in the concrete tower section.

$$g_{11} = D_c = \sum_{i=1}^j \frac{n_{ci}}{N_{ci}} - 1 \leq 0 \quad (18)$$

where n_{ci} and N_{ci} are the number of cycles and fatigue limit of the concrete material, respectively.

2.2.5. Geometric constraints

The diameter at the top of the tower segment should not be less than the diameter at the bottom:

$$g_{12} = r_{st} - r_{sb} \leq 0 \quad (19)$$

$$g_{13} = r_{ct} - r_{cb} \leq 0 \quad (20)$$

The bottom radius of the steel tower segment should not be greater than the top radius of the concrete tower segment:

$$g_{14} = r_{sb} - r_{ct} \leq 0 \quad (21)$$

The thickness of the concrete tower segment must satisfy Eq. (22) [20]:

$$g_{15} = 100 + 0.02r_c - t_c \leq 0 \quad (22)$$

where r_c and t_c are radius and thickness of the concrete tower segment, respectively.

The ratio of the diameter to the thickness for the steel tower segment should not exceed 400 [20]:

$$g_{16} = \frac{2r_s}{t_s} - 400 \leq 0 \quad (23)$$

where r_s and t_s are radius and thickness of the steel tower segment, respectively.

Considering the current height limit requirements for highways and bridges in some countries, it is stipulated that the maximum diameter of steel tower sections shall not exceed 4.5 m [25]. When the thickness of the steel material reaches 40 mm, it becomes difficult to roll the steel tube. It is stipulated that the maximum thickness of the steel tower segment shall not exceed 40 mm. The thickness of the concrete tower segment should be at least 180 mm [20]. In addition, the diameter of the top of the steel tower segment is generally not greater than the width of the nacelle. The range of values for optimization variables are listed in Table 1 and these variables are shown in Fig. 2.

2.3. Objective function

Taking into account factors such as materials, production, transportation, and installation costs, the objective function for the collaborative optimization of the steel-concrete hybrid tower is given by:

$$C = C_c V_c + C_s V_s + C_{pre} V_{pre} + C_f n_s + C_i H_c \quad (24)$$

where C is the objective function; V_i and C_i represent the corresponding volume consumption and comprehensive unit price for each material; C_f represents the cost of a set of connecting flanges; C_t represents the amortization cost of the concrete tower template per unit height; n_s represents the connection number of the steel tower segment; H_c represents the height of the concrete segment. The comprehensive unit price for tower construction costs is listed in Table 2.

To transform the tower optimization design problem with strong constraints into an unconstrained problem, a penalty function is used to consider the constraint conditions in the objective function, resulting in a pseudo objective function:

$$p_i = \begin{cases} 0 & g_i \leq 0 \\ 1 & g_i > 0 \end{cases} \quad (25)$$

$$f = C + 100 \sum_{i=1}^n p_i \quad (26)$$

where p_i is the penalty function; g_i is the value of constraint condition; n is the total number of constraint conditions; and f is the pseudo objective function.

When the structure satisfies all the constraint conditions, the second term in Eq. (26) is 0, and the pseudo-objective function equals the total construction cost of the tower. When there are constraint violations, the second term in Eq. (26) becomes significantly greater than 0, and the pseudo-objective function exceeds the total construction cost of the tower. Therefore, the smaller the value of the pseudo-objective function, the better the tower, as it not only satisfies all the constraint conditions but also achieves cost-effectiveness.

3. Parametric finite element modeling for the hybrid tower

A refined FE model of a steel-concrete hybrid tower based on the open-source program OpenSees is developed to balance computational efficiency and accuracy. Besides, the optimization design of the hybrid tower requires feeding mechanical calculation results back into the

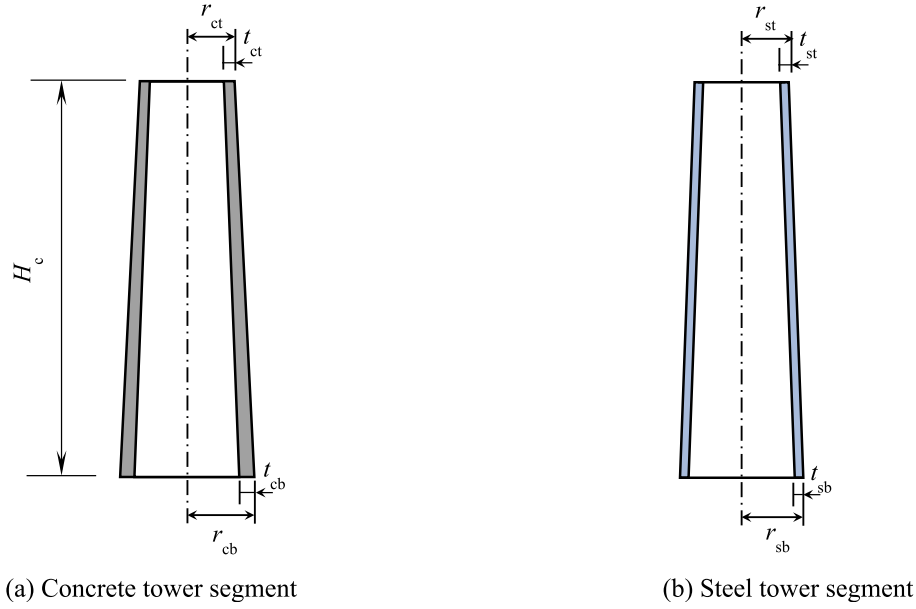


Fig. 2. Design variables of the hybrid tower.

Table 2
Comprehensive unit price for tower construction costs.

Cost items	Symbol	Unit	Unit price
C80 concrete	C_c	yuan /m ³	3300
Steel strands	C_s	yuan /t	10,000
Q345 steel	C_{pre}	yuan /t	11,000
Cost of concrete tower template and installation	C_t	yuan /m	7000
Cost of connecting flanges	C_f	yuan /Set	50,000

optimization algorithm, which then determines the next optimization direction. To facilitate the extensive iterative FE analysis process during optimization, parametric modeling is implemented using the Python language.

3.1. Tower load analysis

The wind turbine tower is in a complex stress state and operating environment. The top of the tower is subjected to loads from blades and nacelle including gravity, aerodynamic loads, deflection forces and moments, gyroscopic forces, and gyroscopic moments. Additionally, the tower also experiences wind loads and its gravity. The loads acting on the top of the tower can be simplified as concentrated forces (F_x, F_y, F_z) and moments (M_x, M_y, M_z) along three coordinate axes, as shown in Fig. 3. The origin of the overall coordinate system is located at the bottom center axis of the hybrid tower. This coordinate system does not rotate with the nacelle. Among these, the x-axis is parallel to the horizontal direction of the wind turbine axis, and the z-axis is vertical.

The wind load on the wind turbine tower is characterized by non-uniform and non-linear distribution due to the effect of wind shear. Along the height of the tower, the wind load increases due to the wind shear. The variation of average wind speed with height can be described by the following exponential function [26]:

$$\frac{v}{v_0} = \left(\frac{z}{z_0} \right)^\alpha \quad (27)$$

where v and z represent the average wind speed and height at any given point, respectively; v_0 and z_0 represent the average wind speed and height at the reference height, respectively; α is the ground wind shear

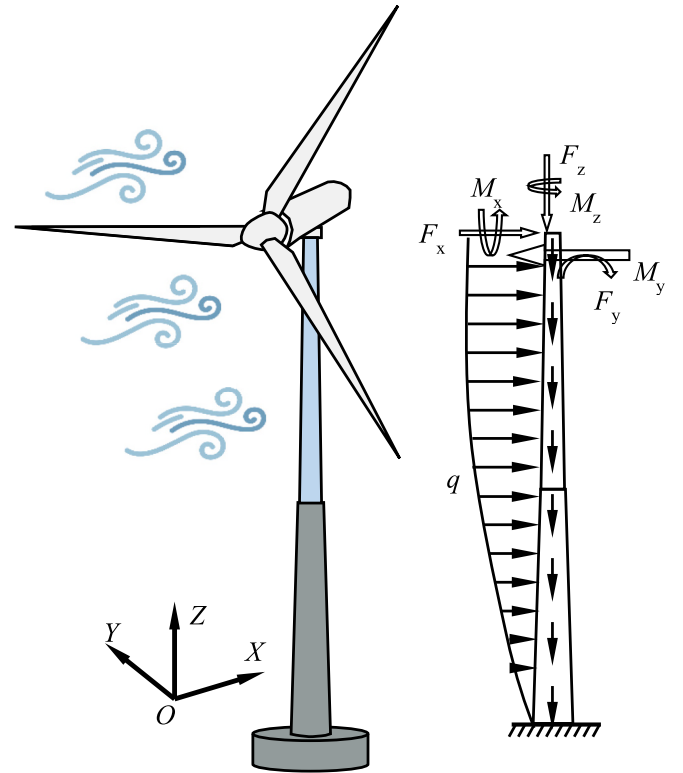


Fig. 3. Loads acting on the hybrid tower.

exponent.

The line load value at any height of the tower body $q(z)$ can be calculated as:

$$q(z) = \frac{1}{2} \rho C_d v(z)^2 D(z) \quad (28)$$

where ρ is the air density, taken as 1.25 kg/m³; C_d is the drag coefficient, taken as 1.0 for a circular structure [27]; $v(z)$ is the incoming wind speed; $D(z)$ is the outer diameter of the tower at any point.

The tower not only experiences aerodynamic loads generated during

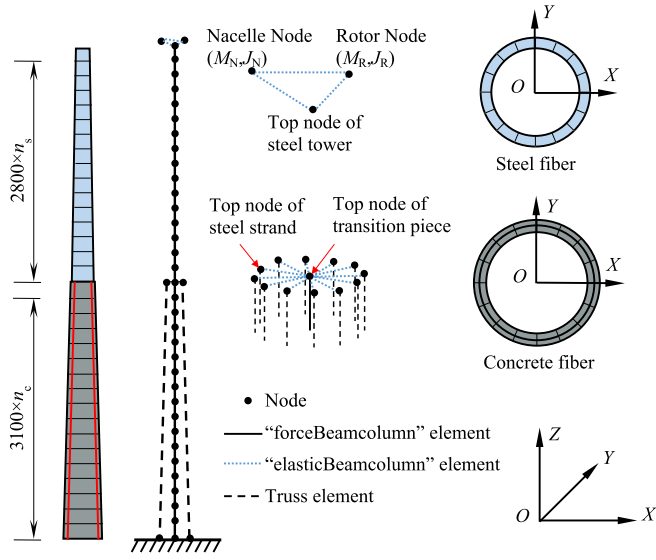


Fig. 4. Finite element model of hybrid tower (unit:mm).

Notes: n_s and n_c represent the number of steel tower sections and the number of prefabricated concrete ring, respectively.

normal operation of the wind turbine for power generation but also encounters loads caused by startup, shutdown, pitch angle changes, and emergency braking. Therefore, IEC 61400-1 defines different load cases and corresponding wind models based on various operating states of the wind turbine [26]. These load cases can be categorized into ultimate loads and fatigue loads, depending on the type of analysis. In this study, both the ultimate and fatigue loads generated at the top of the tower by the wind turbine are provided by the wind turbine manufacturer. The wind loads distributed along the tower body are calculated using Eq. (28). The self-weight of the tower is calculated based on material gravity density. The combination of ultimate loads considered in the analysis includes the structural self-weight, wind loads, and the ultimate loads of the wind turbine.

3.2. Parametric modeling

3.2.1. Adaptive splitting of tower segment

To meet the requirements of transportation and manufacturing processes in practical engineering, the steel tower segment is divided into several sections for welding, and the concrete tower segment is divided into several prefabricated rings for assembly. The outer diameter of each steel tower section varies continuously along the height, with a uniform distribution of thickness, while the outer diameter and thickness of each concrete prefabricated ring vary continuously along the height. Referring to typical steel rolling equipment and concrete formwork size requirements, the heights of the steel tower sections and concrete prefabricated rings in this study are set at 2.8 m and 3.1 m, respectively. Therefore, the steel and concrete tower segment are divided into units with lengths of 2.8 m and 3.1 m along the height direction. These units are simplified using cylindrical sections with uniform cross-sections, and the outer diameter and thickness are taken at the center of each section. For any concrete ring or steel section, the radius and thickness are interpolated based on the top and bottom radii and thickness using Eq. (29) and Eq. (30). Considering formwork requirements, the thickness calculation for concrete segment involves rounding to the nearest integer.

$$r_{ij} = (j + 0.5) / n_i \times (r_{it} - r_{ib}) + r_{ib}, \quad i = c, s \quad (29)$$

$$t_{ij} = \begin{cases} \text{int}[(j + 0.5) / n_i \times (r_{it} - r_{ib}) + r_{ib}], & i = c \\ (j + 0.5) / n_i \times (r_{it} - r_{ib}) + r_{ib}, & i = s \end{cases} \quad (30)$$

where r_{ij} and t_{ij} are the radius and thickness of the j -th section of the i material tower segment. n_i is the number of the sections for i material tower segment.

3.2.2. Element types

The fiber beam-column elements based on the flexibility method is selected for tower segments, with each element having 5 integration points. For concrete tower segment elements, the cross-section is divided into 16 fibers radially and 2 fibers circumferentially, while steel tower segment elements have 16 fibers radially and 1 fiber circumferentially. The transition piece, with a thicker cross-section, is divided into 16 fibers radially and 5 fibers circumferentially. Besides, prestressed steel strands are simulated using truss elements.

3.2.3. Interaction

Due to the significant distribution range of the nacelle and blades, their rotational inertia effects need to be considered when simplifying them into mass points. Previous references [16–18] have not considered the rotational inertia effects of the top nacelle and blades on the structural dynamic characteristics. In this study, the translational mass and rotational inertia of the nacelle (M_N, J_N) and rotor (M_R, J_R) are set using the “mass” command at the corresponding node positions, and they are connected to the top nodes of the steel tower segment using elastic beam-column elements with high stiffness. The distribution mass of the tower body is calculated by multiplying volume and density and is discretized to various nodes. The steel-concrete hybrid tower employs an external prestressing method, and the externally prestressed steel strands are distributed circumferentially on the inner side of the concrete tower segment wall. To simulate the interaction between the steel strands and the concrete tower segment, 12 nodes of truss elements are placed at the top of the transition piece and the bottom of the concrete tower segment, respectively. The top nodes of the steel strands and the top node of the transition piece are connected by weak stiffness elastic beam-column elements to achieve the collaborative deformation of the steel strands with the concrete tower segment, while not affecting the sliding of the steel strands inside the wall due to the deformation of the concrete tower segment [28].

3.2.4. Material constitutive modeling

Since the elastic-plastic behavior of materials is not considered in the design process, steel and concrete are set as elastic materials. The elastic moduli of steel and concrete are 36,000 and 206,000 MPa, respectively. The mass density of concrete is defined as 2500 kg/m³. Considering the impact of anti-corrosion coatings and connecting accessories on the overall quality of the steel tower segment, the mass density of steel is defined as 8500 kg/m³ [29]. The constitutive model for prestressed steel strands employs the Steel02 uniaxial material model with a hardening ratio of 0.01 [28].

3.2.5. Boundary conditions and load application

The bottom nodes of the steel strands and the bottom node of the concrete tower segment are set as fully fixed. By setting the initial stress of the steel strands material and using zero loading, prestress is applied to the concrete. The gravity of the tower is applied to discrete nodes, and its value is the mass of the tower body at the corresponding node multiplied by the gravitational acceleration. To verify the structural responses under three conditions: ultimate, normal operation, and fatigue, loads are applied at the top of the tower segment for each condition. After applying the load for each condition, the structural responses are recorded, and the “remove” command is used to remove the corresponding load condition for the application of the following load condition. Additionally, since the tower segment is a highly flexible structure with significant geometric nonlinearity, second-order P -Delta effects are considered during modeling. Linear geometric transformations of beam stiffness and resistance from the basic coordinate

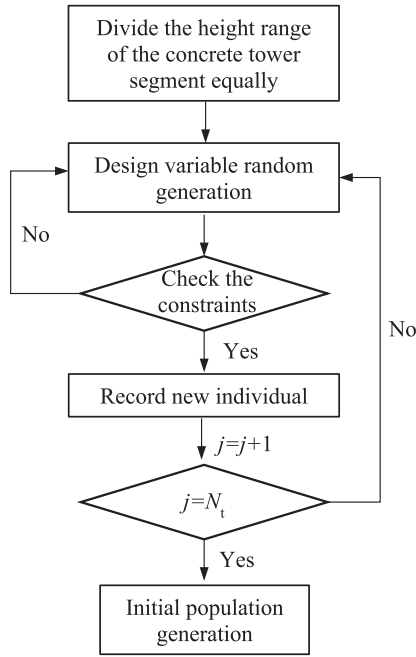


Fig. 5. Flowchart of initial population optimization strategy.

system to the global coordinate system are executed. The overall finite element model is illustrated in Fig. 4.

4. Intelligent optimization algorithms

Evolutionary algorithm is an adaptive stochastic global search and optimization method formed by simulating the characteristics of population competition, natural selection, genetic inheritance and mutation in biological evolution through computer simulation. Genetic algorithm (GA) and differential evolution (DE) algorithm are both classic evolutionary algorithms with strong global convergence ability and robustness [30–32]. Both algorithms involve mutation, crossover, and selection operations but differ in each operator and evolutionary process [33]. Considering the nonlinearity, non-convexity, multimodality, and discontinuity of hybrid tower optimization problems, this paper proposes appropriate improvements based on standard GA and DE algorithms and develops modified genetic algorithms and differential evolution algorithms suitable for hybrid tower optimization.

4.1. Initial population optimization strategy

The traditional evolutionary algorithms generally employ a random initialization approach for the population. If the initial solutions are close to the optimal solution, it can accelerate the search for the optimal solution. However, during the generation of the initial population, the lack of directionality often results in new individuals being generated far from the optimal solution. Additionally, these new individuals may fail to satisfy geometric constraints, such as the top radius of the tower segment greater than the bottom radius. This increases the number of

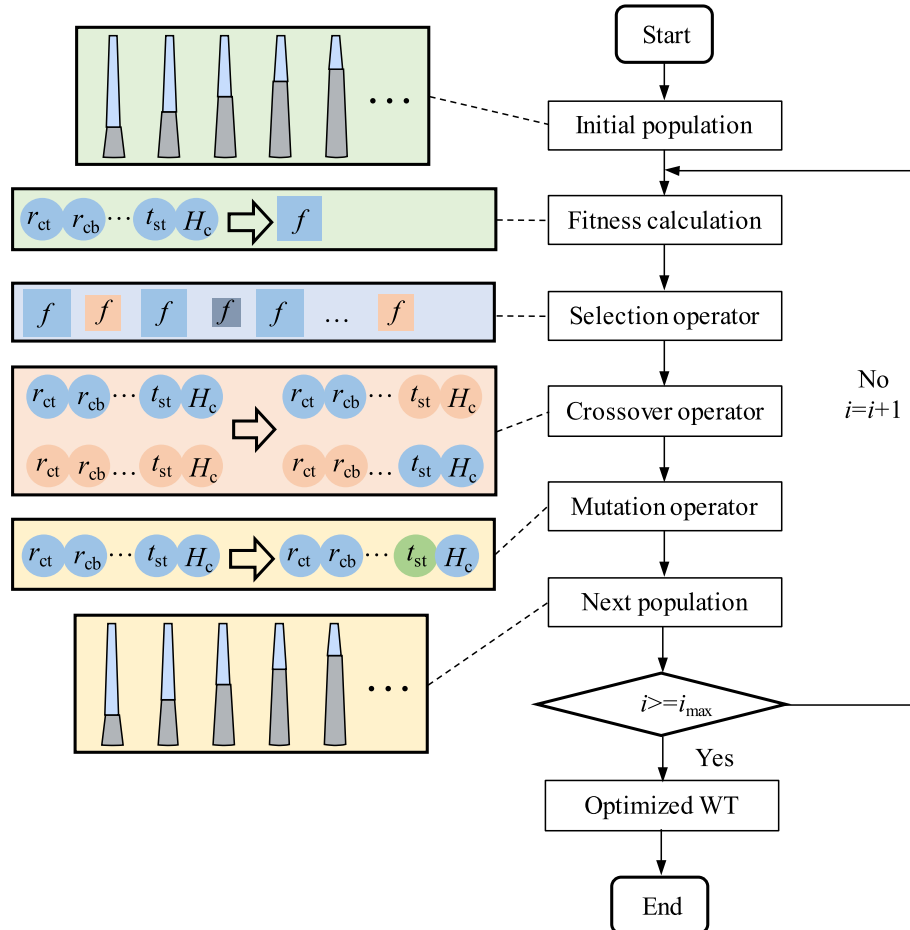


Fig. 6. Modified genetic algorithm flowchart.

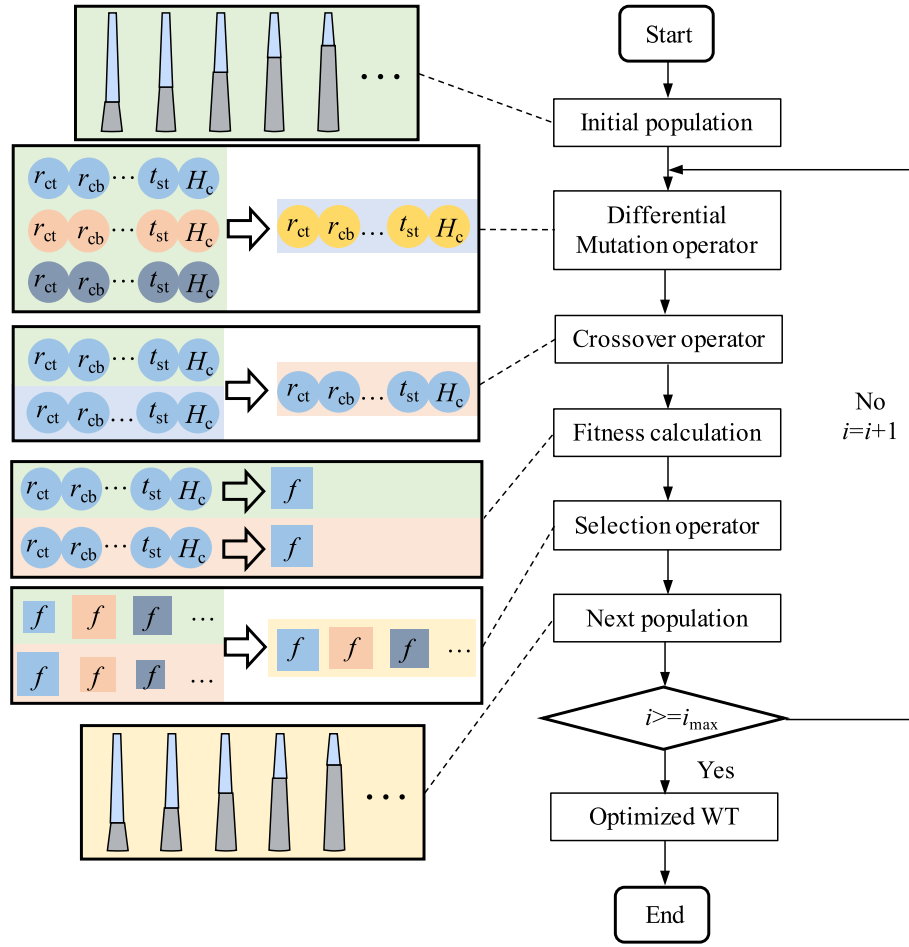


Fig. 7. Modified differential evolution algorithm flowchart.

ineffective individuals (those violating constraints), leading to additional computational time in finite element analysis.

In order to ensure a wide distribution of individuals in the initial population, this study first determines the range of distribution for the height of each concrete tower segment based on the principle of uniform distribution. Subsequently, other design parameters are randomly generated. To address the issue of new individuals not satisfying geometric constraints, a constraint-based strategy is employed to improve the situation. When a new individual is generated, the relevant verification of its design variables is performed through Eqs. (19–23). If the requirements are met, the new individual is considered valid; otherwise, the same operation is repeated until the constraints are satisfied. Until the number of effective individuals generated reaches the population size N_t , the initial population is generated. The flowchart of the initial population optimization strategy is shown in Fig. 5.

4.2. Fitness calculation method based on hierarchical division

Directly transforming the objective function to calculate the individual fitness may introduce scaling issues. When the fitness is too small, it may lead to slow convergence in the search, while excessively amplifying the fitness of dominant individuals can result in rapid convergence. This paper addresses this issue using a rank-based fitness assignment approach [34]. The individuals are first ranked based on their objective function values in descending order, and then their fitness is determined according to their position in the ranking. The fitness F_i of the i -th individual is calculated as follows:

$$F_i = 2 - SP + 2(SP - 1) \frac{i - 1}{N_t - 1} \quad (31)$$

Where N_t is the population size; i is the position of the individual in the population, $i = 1, 2, \dots, N_t$; SP denotes the selection pressure, set to 1.8 [34].

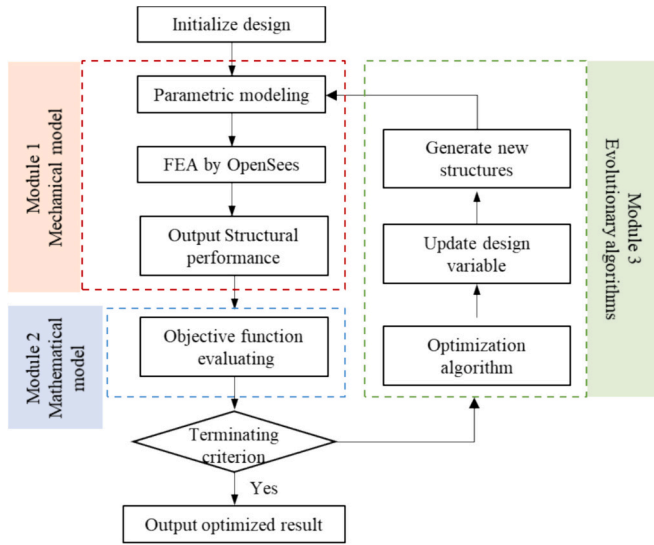
4.3. Modified genetic algorithm

The process for the modified genetic algorithm is as follows: 1) Real-number encoding rules and the initial population optimization strategy in Section 4.1 are utilized to generate the initial population. 2) Individuals in the initial population (or new population) are decoded to conduct FE analysis. The pseudo-objective function and the fitness values of individuals in the population are calculated using Eq. (26) and Eq. (31), respectively. 3) Roulette wheel selection is adopted to choose some excellent individuals as parents based on their fitness values [35]. 4) Simulated binary crossover is employed to generate offspring by crossing individuals in the mating pool with a certain probability p_c [36]. 5) Polynomial mutation method is used to perform a certain probability p_m of mutation on individuals in offspring, ultimately obtaining a new generation population [37]. Repeat steps (2–5) to generate a new generation population. When the iteration reaches the set maximum iteration number, the optimization process terminates, and the optimal solution is output. The modified genetic algorithm flowchart for optimization design of hybrid tower is shown in Fig. 6.

Table 3

Main Parameter setting of algorithms.

Algorithm	Pop_size (N_t)	Iter_max (i_{max})	Crossover probability (p_c)	Mutation probability (p_m/F)
GA	30,50,80	100	0.8	0.5
DE	30,50,80	100	0.8	0.01

**Fig. 8.** Flowchart of automatic optimization design framework.

4.4. Modified differential evolution algorithm

The process for the modified differential evolution algorithm is as follows: 1) Real-number encoding rules and the initial population optimization strategy in Section 4.1 are utilized to generate the initial population. 2) Differential mutation operations are performed on parent individuals in the initial population (or new population) to generate mutated offspring [38]. 3) Binomial crossover between mutated offspring and parent individuals generates crossed offspring with a

certain probability p_c [39]. 4) Parent and crossed offspring individuals are both decoded to conduct FE analysis. The pseudo-objective function and the fitness values of individuals in the population are calculated using Eq. (26) and Eq. (31), respectively. 5) A Greedy selection mechanism is adopted to select excellent individuals from both parent and crossed offspring based on their fitness values forming a new population [33]. The modified differential evolution flowchart for the optimization design of the hybrid tower is shown in Fig. 7. The main parameter settings of the two algorithms are listed in Table 3.

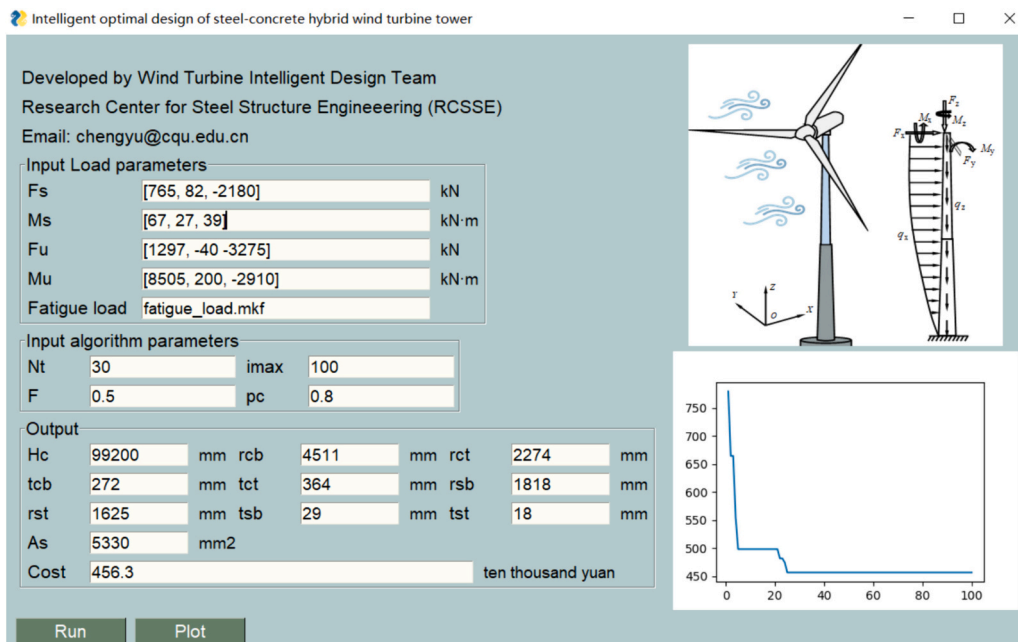
4.5. Automatic optimization framework

Fig. 8 illustrates the automatic optimization design framework of a hybrid tower. This framework mainly consists of three modules: mechanical analysis module (Module 1), objective function calculation module (Module 2), and evolutionary algorithm module (Module 3). Module 1 receives input design variables based on OpenSees for automated modeling and FE analysis of hybrid towers and outputs structural responses to Module 2. Module 2 receives the structural response output from Module 1 for constraint evaluation and objective function calculation. Module 3 receives the objective function output from Module 1 for fitness calculation and evolutionary operations, thereby generating a new generation of population and transmitting it to Module 1. Based on Python secondary development technology, an interface has been developed to connect optimization mathematical models, structural analysis models, and intelligent optimization algorithm modules, thereby achieving automatic optimization design of a hybrid tower that meets design constraints.

Based on the automatic optimization design framework mentioned above, an interactive and user-friendly graphical user interface (GUI) has been established for intelligent optimization design of hybrid towers. After inputting load and algorithm parameters, click the “Run” button to display the optimized total construction cost and various design variables, as shown in Fig. 9.

5. Case study

The intelligent optimization design method proposed in this paper was applied to design a 160 m steel-concrete hybrid tower, and the

**Fig. 9.** Graphical user interface for automatic optimization.

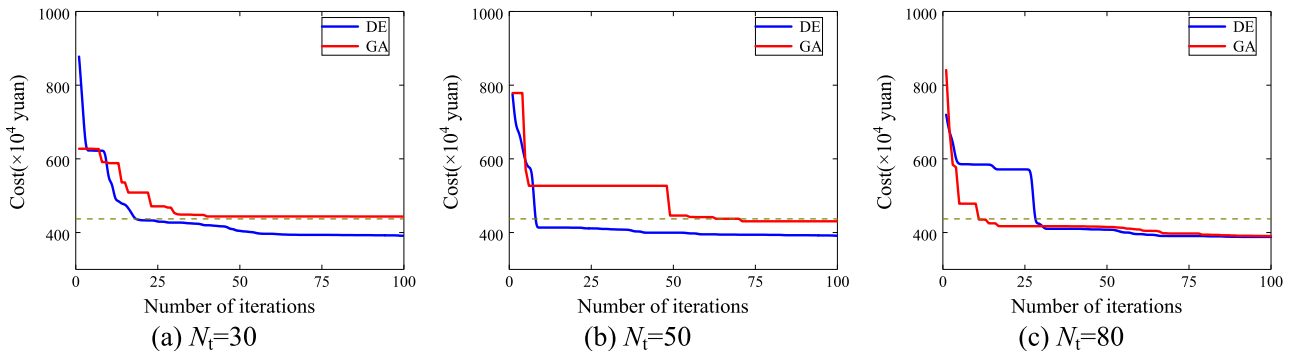


Fig. 10. Cost convergence curves of different population size.

Table 4
Optimization results of different algorithms.

N_t	DE			GA		
	Optimal solution ($\times 10^4$ yuan)	Optimization amplitude (%)	Computing time(s)	Optimal solution ($\times 10^4$ yuan)	Optimization amplitude (%)	Computing time(s)
30	391.3	10.5	9371	443.6	0	9477
50	391.1	10.5	14,844	430.7	1.4	17,829
80	388.2	11.2	28,262	390.5	10.6	24,193

Table 5
Design variables for hybrid tower structure.

Design variable parameters	Symbols	Manual optimization	Intelligent optimization
Height of concrete tower segment (mm)	H_c	105,400	102,300
Radius at concrete tower segment base (mm)	r_{cb}	5010	4881
Radius at concrete tower segment top (mm)	r_{ct}	2400	2486
Thickness at concrete tower segment base (mm)	t_{cb}	230	198
Thickness at concrete tower segment top (mm)	t_{ct}	340	246
Radius at steel tower segment base (mm)	r_{sb}	2325	1986
Radius at steel tower segment top (mm)	r_{st}	1843	1500
Thickness at steel tower segment base (mm)	t_{sb}	18	23
Thickness at steel tower segment top (mm)	t_{st}	11	11
Area of prestressed steel strands (mm^2)	A_s	4900	5386

results were compared with ones obtained through conventional manual modeling optimization to validate the effectiveness of this method.

5.1. Wind turbine parameters

The selected tower case was constructed in a wind farm in China. The rated power of the wind turbine is 5.0 MW, with a rated wind speed of 11 m/s and a cut-off wind speed of 25 m/s. The rotor diameter is 193 m, and the rated blade speed is 5.482 rpm. The total mass of the nacelle, hub, and blades is 235 tons, and the equivalent center of mass is located 4 m from the windward side of the tower axis. The nacelle dimensions are 12 m $\times$ 4.5 m $\times$ 4.0 m. The average wind speed in this wind farm ranges from 5.9 to 6.8 m/s, with a 50-year maximum wind speed of 32.11 m/s. The ground wind shear exponent α is 0.2. The concrete is made of C80 concrete, the steel is made of Q355 steel, and the steel strand is made of high-strength steel strand with an ultimate strength standard value of 1860 MPa. The design strength of the material is

determined by Chinese code GB 50010–2010 [40].

5.2. Optimal results

5.2.1. Objective function

The relationship between the construction cost of the steel-concrete hybrid tower and the number of iterations is depicted in Fig. 10. As seen, the cost decreases rapidly in the early stages of optimization, and in the later stages, and gradually converges in the later stage of optimization. Due to the random generation of individuals in the initial population of the optimization algorithm, design constraints were not fully satisfied in the early stages of optimization, resulting in higher optimization results for both algorithms than conventional manual optimization design. The optimization results of different algorithms are listed in Table 4. With an increase in the number of iterations and population size, both types of intelligent optimization algorithms effectively reduce the construction cost of the hybrid tower by approximately 11%. Among them, the DE algorithm exhibits higher optimization efficiency than GA. Even with a population size of 30, satisfactory results can be obtained after 100 iterations. GA produces optimization results comparable to conventional manual optimization design with a population size of 30 and 50. When the population size of GA is increased to 80, it achieves results similar to DE. Additionally, conventional manual optimization design currently requires experienced structural engineers to spend approximately two days on model adjustments and optimization. In contrast, intelligent optimization design method takes only about four hours. Adopting intelligent optimization design can significantly enhance design efficiency and reduce construction costs. DE algorithm is suggested for optimizing the hybrid tower to improve optimization efficiency.

5.2.2. Design variables

The main design parameters of the steel-concrete hybrid tower between intelligent optimization and manual optimization are compared in Table 5. The variation curves of cross-section diameter and thickness along tower height for two optimization design methods are illustrated in Fig. 11. The height of the concrete segment is a crucial parameter. Conventional manual optimization of this parameter would consume a significant amount of time due to the following reasons: 1) The natural frequency of the hybrid tower does not monotonically increase with the increase of the concrete segment [30]; 2) As the height of the concrete tower segment increases, leading to higher vertical compressive stresses

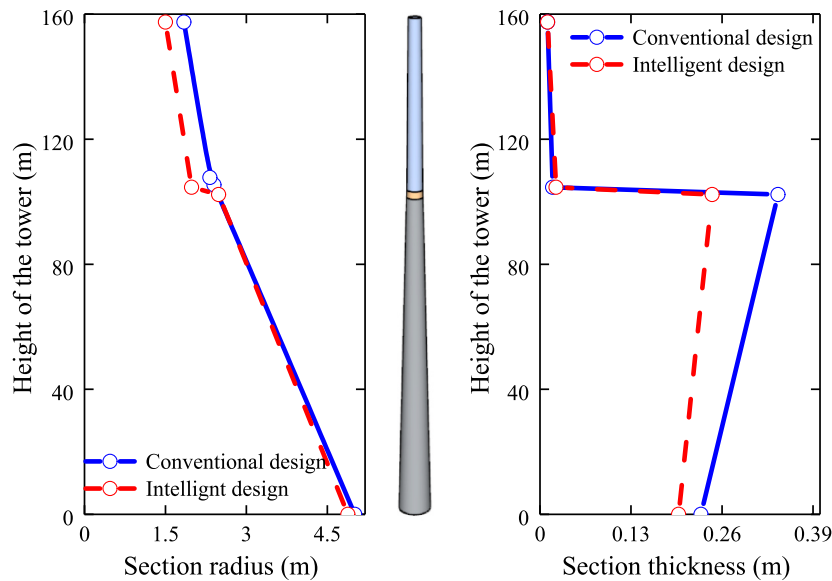


Fig. 11. Comparison of intelligent and manual optimization design.

induced by gravity, reducing the required prestressed steel strands area to resist the vertical compressive stress generated by bending moments; 3) With the increase in the height of the concrete tower segment, the amortization cost of the formwork will also increase. The radius at the concrete tower segment base has also decreased, contributing to a reduction in foundation costs. It is worth mentioning that not every design variable has decreased after intelligent optimization. For instance, the thickness at the bottom of the steel tower segment has increased by 27.7% after intelligent optimization. This is considered reasonable since the bottom radius of the steel tower has decreased by 14.6%, and the height of the steel tower segment has slightly increased.

5.2.3. Design constraints

The evolution history of design constraints during the optimization process is recorded and illustrated in Fig. 12. It can be observed that constraints such as tensile stress in concrete under serviceability limit state, strength of the steel tower segment, tower top displacement, rotation angle, and local stability of the steel tower are easily satisfied. However, constraints such as natural frequency, tensile and compressive stress in concrete under ultimate limit state, concrete fatigue damage, and overall stability of the steel tower act as controlling constraints during the optimization process. In the later stages of optimization, the worst individuals in the population still fail to satisfy these constraints, while the optimal individuals are close to the limit values.

5.2.4. Objective function

The internal forces along the height of the tower between intelligent optimization design and manual optimization design are compared in Fig. 13. From Fig. 13 (a), it can be seen that the axial force undergoes a sudden change at the top of the transition piece due to the application of prestress. Although the radius and thickness of the intelligent optimized concrete tower segment are reduced, the prestressed steel strands area is increased to prevent concrete from undergoing tensile cracking. Therefore, the axial force in intelligently optimized towers is larger than in conventional design. In Fig. 13 (b), it is evident that the shear force in the intelligent optimized tower decreases as a result of the reduced wind load on the tower due to the decreased radius and thickness of the concrete tower segment. Fig. 13 (c) indicates that the change in bending moment is not significant for both designs.

The stress and fatigue strength variation curves along the height of the tower between intelligent optimization design and conventional optimization design are compared in Fig. 14. Compared with traditional

optimization design, under the premise of meeting design constraints, the stress and fatigue damage of the tower in each section of intelligent optimization design are higher than those in traditional design, and the stress or fatigue damage of all sections are close to the limit values. The compression utilization rate of the concrete tower segment exceeds 80%, indicating that the optimized tower fully utilizes the material advantages of concrete under compression.

6. Conclusions

In this study, an integrated intelligent optimization design framework is proposed based on open-source FE program OpenSees and modified evolutionary algorithms for steel-concrete hybrid wind turbine towers. The following major findings are offered:

- 1) An optimization mathematical model for the hybrid tower was established, introducing the comprehensive design guidelines and presenting the penalty function form for handling effectively constraint conditions.
- 2) A parametric FE model of a hybrid tower was developed using OpenSees, considering the mass characteristics of the top wind turbine, the interaction between prestressed steel strands and concrete, and the influence of geometric nonlinearity.
- 3) Based on the standard evolutionary algorithm, an efficient global optimization algorithm suitable for hybrid tower optimization was developed by introducing an initial population optimization strategy and a fitness calculation method based on hierarchical division.
- 4) The case study results demonstrate that the total construction cost of intelligent optimization design is reduced by about 11% compared with the conventional manual optimization design while satisfying all design constraints. Among them, the modified differential evolution algorithm has higher optimization efficiency than the modified genetic algorithm.
- 5) The optimization results show that the total construction cost is lowest when the height of the prestressed concrete tower segment accounts for about 65% of the total tower height.

CRediT authorship contribution statement

Yu Cheng: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Yannan Zhao:** Writing – original draft, Resources, Methodology, Investigation, Conceptualization. **Hongtuo Qi:** Writing – review

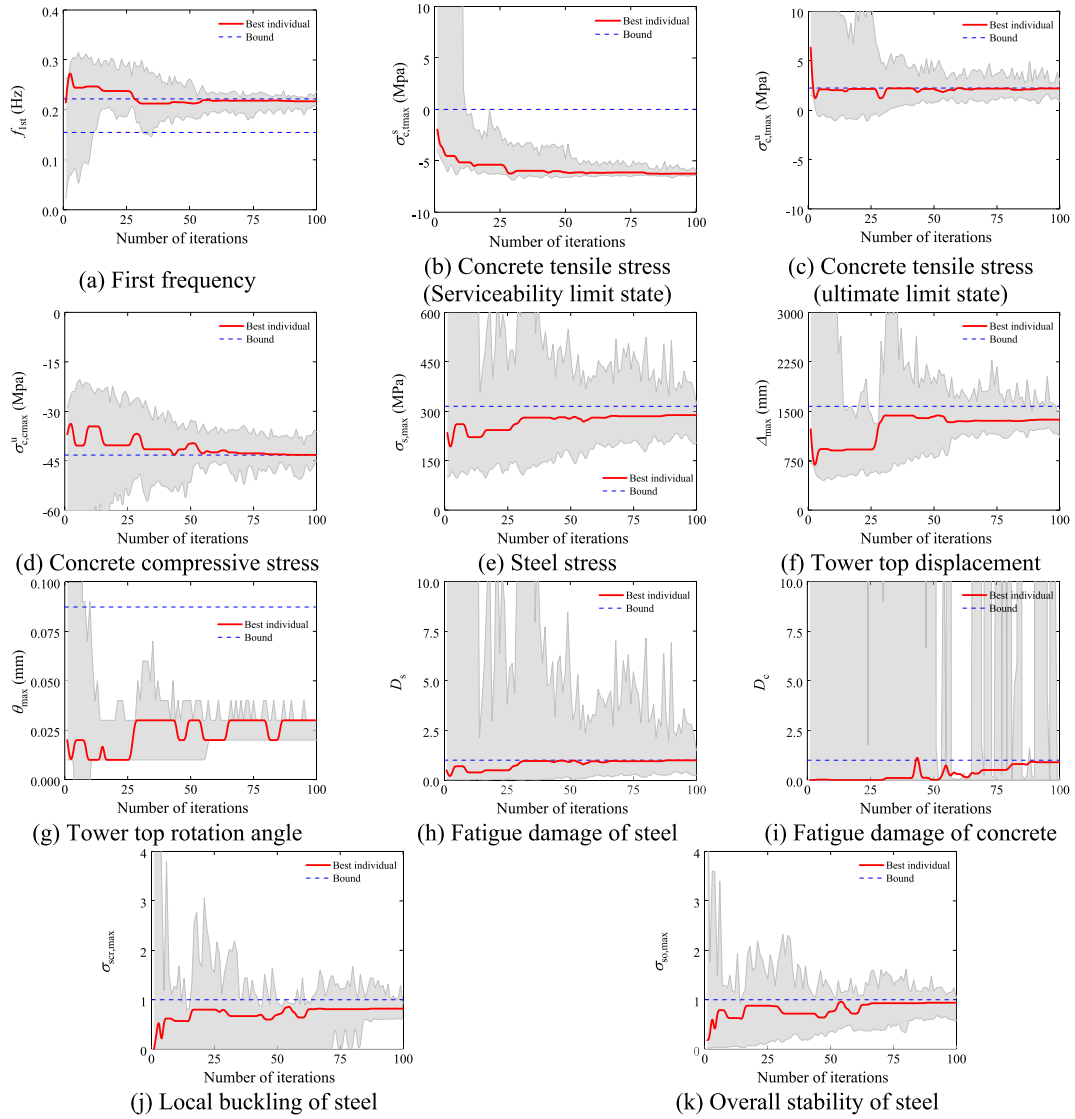


Fig. 12. History of the optimization constraint.

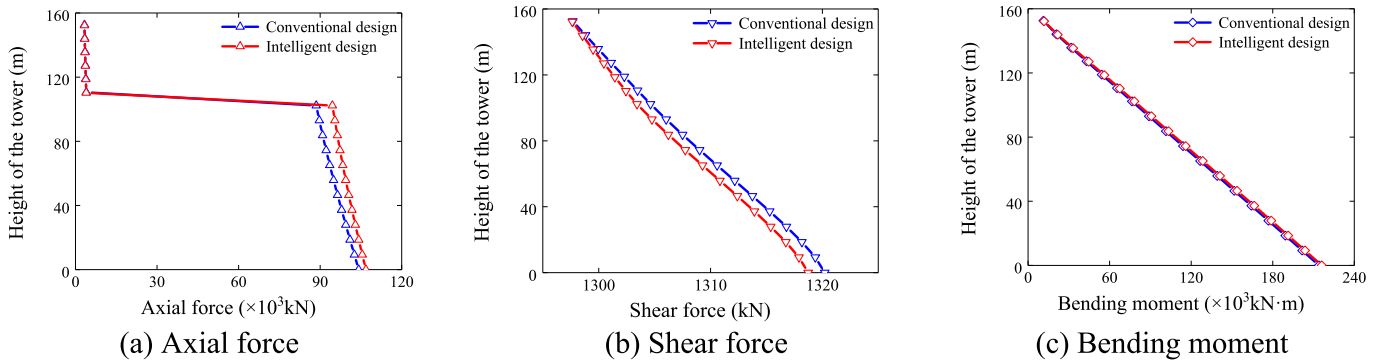


Fig. 13. Internal forces along the tower height.

& editing, Supervision, Resources, Investigation, Conceptualization.
Xuhong Zhou: Validation, Supervision, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

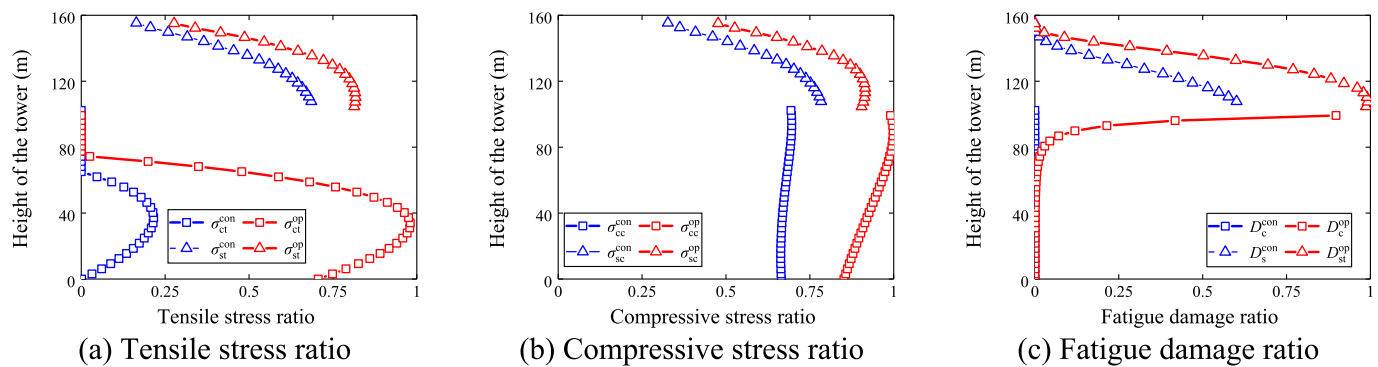


Fig. 14. Stress and fatigue damage along the tower height.

Data availability

Data will be made available on request.

Acknowledgments

This research is supported by the National Key Research and Development Program of China (Grant No. 2022YFC38017), the National Natural Science Foundation of China (Grant No. 52378128), Graduate Scientific Research and Innovation Foundation of Chongqing, China (CYB22028), the National Key Research and Development Program of Municipality of Chongqing (Grant No. CSTB2022TIAD-KPX0140), and Chongqing Academicians Leading the Special Project for Science and Technology Innovation Guidance China (Grant No. 2022YSZX-JCX0009CSTB).

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